

NEUROLENS: A MULTI-AXIS XAI BENCHMARK FOR BRAIN TUMOR CLASSIFICATION

FINAL PROJECT REPORT

Mounia Abdelmoumni & Adnane Erekraken

MSc Artificial Intelligence, CentraleSupélec (Paris-Saclay)

{mounia.abdelmoumni, adnane.erekraken}@student-cs.fr

ABSTRACT

Explainable AI (XAI) methods are essential for deploying deep learning in clinical settings such as brain tumor classification, yet there is no consensus on which explanation method to trust. We present NEUROLENS, a systematic benchmark that evaluates eight attribution methods from six families; Grad-CAM, EigenCAM, Integrated Gradients, SmoothGrad, DeepLIFT, Occlusion, LIME, and Attention Rollout; across four architectures (ResNet50, DenseNet121, EfficientNetV2-S, ViT-B/16) on a 13,351-image brain MRI dataset. Unlike prior work that relies on qualitative comparison alone, we assess explanations along six quantitative axes: (1) faithfulness via deletion/insertion AUC with bootstrap confidence intervals and Wilcoxon signed-rank tests ($N=100$); (2) model dependence via cascading weight randomization; (3) concept specificity via class-vs-class TCAV with statistical significance testing; (4) stability under Gaussian input perturbation; (5) inter-method agreement via Spearman rank correlation; and (6) computational cost. Our results reveal fundamental trade-offs: Grad-CAM is the most faithful and stable method on CNNs but is architecture-dependent (sanity-check SSIM = 1.0), while SmoothGrad is the only partially model-dependent method (SSIM = 0.56) but the least stable. We recommend presenting both methods side-by-side for clinical deployment and identify Grad-CAM as the sole Pareto-optimal method on the cost–faithfulness frontier.

1 INTRODUCTION AND MOTIVATION

Deep learning models achieve near-human accuracy on medical image classification tasks, including brain tumor classification from MRI scans. However, clinical adoption requires more than accuracy: clinicians need to understand *why* a model predicts a given diagnosis before trusting it. Explainable AI (XAI) methods promise to surface this reasoning by producing attribution maps that highlight which input regions drive the prediction.

The proliferation of XAI methods—gradient-based, perturbation-based, attention-based, concept-based—creates a selection problem: different methods produce visibly different explanations for the same image, and there is no consensus on which to trust. Prior benchmarks (Samek et al., 2021) typically compare methods on natural-image datasets (e.g., ImageNet), which may not transfer to the clinical domain where spatial precision, stability across scans, and model dependence are critical.

We address the following research questions:

- **RQ1—Faithfulness:** Do attributions reflect the features the model actually relies on?
- **RQ2—Model dependence:** Are explanations functions of learned weights, or of architecture?
- **RQ3—Concept specificity:** Do models use class-specific features, or are there shortcuts?
- **RQ4—Stability:** Are explanations robust to small input perturbations?
- **RQ5—Agreement:** Do different methods agree on which pixels matter?
- **RQ6—Cost:** What is the Pareto frontier of quality vs. compute?

Potential applications include: clinical decision support for radiologists, regulatory auditing of medical AI systems, and principled method selection for XAI practitioners deploying models on safety-critical imaging tasks.

2 PROBLEM DEFINITION

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$ be a neural network parameterized by θ , mapping an input image $x \in \mathcal{X} \subset \mathbb{R}^{C \times H \times W}$ to a vector of K class logits. An *attribution method* $\mathcal{A}$ produces a saliency map $s = \mathcal{A}(f_\theta, x, k) \in \mathbb{R}^{H \times W}$ for target class k , where s_{ij} indicates the importance of pixel (i, j) to the prediction.

The core challenge is that there is no ground truth for s —we do not know the “correct” explanation. This makes evaluation inherently indirect. We adopt six proxy evaluation axes, each formalizing a desirable property of a faithful attribution:

Faithfulness (deletion/insertion). Let π be the permutation that sorts pixels by decreasing s_{ij} . Define $x_r^{(\text{del})}$ as the image with the top- r fraction of pixels replaced by a blurred baseline, and $x_r^{(\text{ins})}$ as the blurred image with only the top- r fraction of original pixels restored. A faithful method should yield low deletion AUC: $\text{AUC}_{\text{del}} = \int_0^1 f_\theta(x_r^{(\text{del})})_k dr$, and high insertion AUC: $\text{AUC}_{\text{ins}} = \int_0^1 f_\theta(x_r^{(\text{ins})})_k dr$. We report the gap $\Delta = \text{AUC}_{\text{ins}} - \text{AUC}_{\text{del}}$.

Model dependence (cascading randomization). Following Adebayo et al. (2018), we progressively randomize the weights of f_θ from the output layer to the input layer and measure how the attribution changes: $\text{SSIM}(\mathcal{A}(f_\theta, x, k), \mathcal{A}(f_{\theta'}, x, k))$, where θ' denotes the partially randomized parameters. A faithful method should show decreasing SSIM as more layers are randomized.

Concept specificity (TCAV). Following Kim et al. (2018), for a concept C with learned activation vector v_C , the TCAV score is $\text{TCAV}_{C,k} = \frac{1}{|\mathcal{X}_k|} \sum_{x \in \mathcal{X}_k} \mathbf{1}[\nabla_h f_\theta(x)_k \cdot v_C > 0]$, where h denotes the penultimate-layer activations.

3 RELATED WORK

Gradient-based attributions. Grad-CAM (Selvaraju et al., 2017) produces class-discriminative localization maps by weighting feature maps with gradient-derived importance. Integrated Gradients (Sundararajan et al., 2017) provides an axiomatic framework satisfying sensitivity and implementation invariance. SmoothGrad (Smilkov et al., 2017) reduces gradient noise by averaging over Gaussian-perturbed copies.

Propagation-based methods. DeepLIFT (Shrikumar et al., 2017) assigns attributions by comparing activations to a reference. While related to LRP (Bach et al., 2015), we note that they are distinct methods; we use DeepLIFT via Captum and reserve LRP (e.g., via Zennit) as future work.

Perturbation-based methods. LIME (Ribeiro et al., 2016) fits a local linear surrogate on superpixel perturbations. Occlusion sensitivity (Zeiler & Fergus, 2014) masks image regions and measures prediction change. Both are model-agnostic but computationally expensive.

Attention-based methods. Attention Rollout (Abnar & Zuidema, 2020) aggregates attention weights across transformer layers by iterative matrix multiplication, yielding a global attention map from the CLS token.

Concept-based methods. TCAV (Kim et al., 2018) tests whether predictions are sensitive to human-defined concept directions in activation space. We extend class-vs-class TCAV with a statistical significance test using random-CAV null distributions.

Evaluation frameworks. Adebayo et al. (2018) introduced cascading randomization to test model dependence. Pet-siuk et al. (2018) proposed deletion/insertion curves. Alvarez-Melis & Jaakkola (2018) studied stability under input perturbation. Our contribution is to combine all six axes in a single benchmark on a medical imaging task, with rigorous statistical testing.

4 METHODOLOGY

4.1 DATASET AND PREPROCESSING

We use the Brain MRI dataset (Sakin, 2023) from Kaggle (13,351 images, 4 classes: glioma, meningioma, pituitary, notumor). Images are resized to 224×224 and normalized with ImageNet statistics. We apply a stratified 80/20 train/val split (10,680/2,671) with light augmentation (horizontal flip, rotation $\pm 15^\circ$, color jitter).

Limitation. The dataset lacks radiologist-annotated segmentation masks, precluding pointing-game or IoU localization metrics. We rely on faithfulness proxies that do not require ground-truth localization.

4.2 MODELS

We fine-tune four pretrained architectures, freezing early layers and training only the last block + classifier head for 7 epochs with AdamW and cosine annealing:

Table 1: Model architectures and validation accuracy.

Model	Family	Trainable/Total params	Val acc.
ResNet50	Residual CNN	15.0M / 23.5M (63.7%)	97.5%
DenseNet121	Dense CNN	—	97.5%
EfficientNetV2-S	Compound CNN	—	98.2%
ViT-B/16	Transformer	14.2M / 85.8M (16.5%)	97.3%

4.3 ATTRIBUTION METHODS

We implement eight methods spanning six families (Table 2). All CNN methods use the last convolutional block as the target layer. For ViT, Grad-CAM uses `encoder.layers[-1].ln_2` (post-MLP LayerNorm) with a reshape transform that drops the CLS token and reshapes 196 patch tokens to a 14×14 grid.

Table 2: Attribution methods implemented in NeuroLens.

Family	Method	Reference
Activation-based	Grad-CAM	Selvaraju et al. (2017)
	EigenCAM	Muhammad & Yeasin (2020)
Backpropagation	Integrated Gradients	Sundararajan et al. (2017)
	SmoothGrad	Smilkov et al. (2017)
Perturbation	DeepLIFT	Shrikumar et al. (2017)
	Occlusion	Zeiler & Fergus (2014)
	LIME	Ribeiro et al. (2016)
Transformer	Attention Rollout	Abnar & Zuidema (2020)
<i>Additionally: k-NN exemplar-based and class-vs-class TCAV</i>		

4.4 EVALUATION PROTOCOL

Faithfulness (RQ1). We compute deletion/insertion AUC on $N=100$ class-balanced validation images for both the best (EfficientNetV2-S) and worst (ViT-B/16) models. We report bootstrap 95% CIs ($n_{\text{boot}}=1000$) and pairwise Wilcoxon signed-rank tests with Bonferroni correction.

Cascading randomization (RQ2). Following Adebayo et al. (2018), we progressively randomize ResNet50 layers from `fc`→`layer4`→`layer3`→`layer2`→`layer1`→`conv1` and measure SSIM and $|\rho|$ (Spearman) between randomized and trained attributions, averaged over 5 images. DeepLIFT was excluded due to a Captum incompatibility with ResNet’s reused ReLU modules.

TCAV (RQ3). For class-vs-class TCAV, we train a linear SVM (SGD hinge loss) to separate class i embeddings from all other classes, obtaining a concept activation vector (CAV). We compute the TCAV score using *gradients* of the predicted logit w.r.t. penultimate activations (correcting a common implementation error that uses activations instead). Statistical significance is assessed with 20 random-CAV null models via a two-sided t -test ($\alpha=0.05$).

Stability (RQ4). For 20 images, we add Gaussian noise ($\sigma=0.05$) three times and measure mean SSIM between clean and perturbed attribution maps.

Agreement (RQ5). For 20 images, we compute pairwise Spearman ρ between pixel-level attributions from each pair of methods.

Cost (RQ6). We profile wall-clock time per explanation on an NVIDIA RTX 4050 Ti (4.3 GB VRAM) and plot the Pareto frontier against faithfulness.

5 EVALUATION RESULTS

Figure 1 provides a qualitative overview of Grad-CAM and SmoothGrad across all four models and tumor classes before we turn to quantitative evaluation.

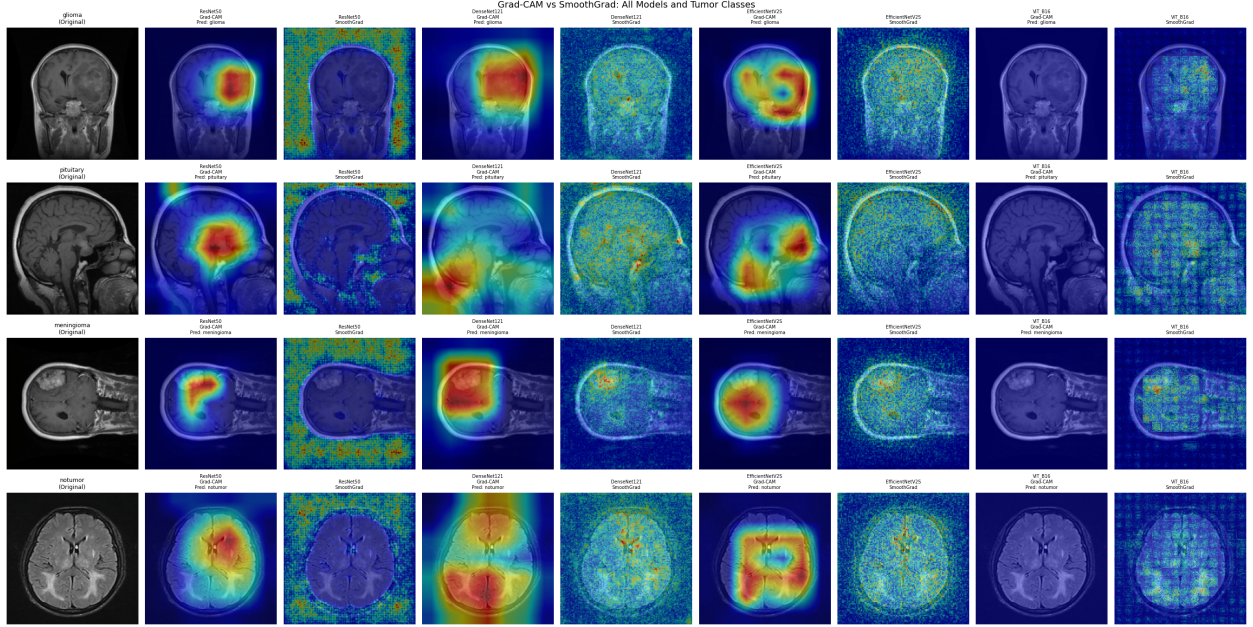


Figure 1: Grad-CAM (left of each pair) vs. SmoothGrad (right) across all models and tumor classes. Grad-CAM produces smooth, spatially coherent heatmaps on CNNs but diffuse maps on ViT. SmoothGrad produces finer-grained, noisier attributions across all architectures. All models predict correctly on these samples.

Two patterns are immediately visible. First, Grad-CAM heatmaps are spatially coherent on CNNs (ResNet50, DenseNet121, EfficientNetV2-S) but become diffuse on ViT-B/16, foreshadowing the faithfulness drop we quantify in §5.1. Second, SmoothGrad consistently highlights fine-grained edge and texture features regardless of architecture, but with visible noise—consistent with the low stability we measure in §5.4. These qualitative observations motivate the six-axis quantitative evaluation that follows.

5.1 FAITHFULNESS (RQ1)

Table 3 shows the faithfulness gap ($\Delta = \text{AUC}_{\text{ins}} - \text{AUC}_{\text{del}}$) with bootstrap 95% CIs. On EfficientNetV2-S, Grad-CAM achieves the highest gap (+0.364), followed by EigenCAM (+0.362), SmoothGrad (+0.322), and Occlusion (+0.308). These four are statistically indistinguishable (Wilcoxon $p > 0.05$ after Bonferroni). IntGrad (+0.113) and DeepLIFT (−0.077) are significantly worse ($p < 0.001$).

On ViT-B/16, the ranking reverses: SmoothGrad becomes the most faithful (+0.359), while Grad-CAM drops to +0.015, indicating strong architecture dependence of CAM methods.

Table 3: Faithfulness gap (Ins. AUC − Del. AUC) on $N=100$ images. Higher is better. Best model: EfficientNetV2-S; worst: ViT-B/16.

Method	EfficientNetV2-S	ViT-B/16
Grad-CAM	+0.364 [0.33, 0.40]	+0.015
EigenCAM	+0.362 [0.33, 0.40]	+0.168
SmoothGrad	+0.322 [0.29, 0.36]	+0.359
Occlusion	+0.308 [0.26, 0.35]	+0.280
IntGrad	+0.113 [0.07, 0.15]	+0.214
DeepLIFT	−0.077 [−0.12, −0.04]	—

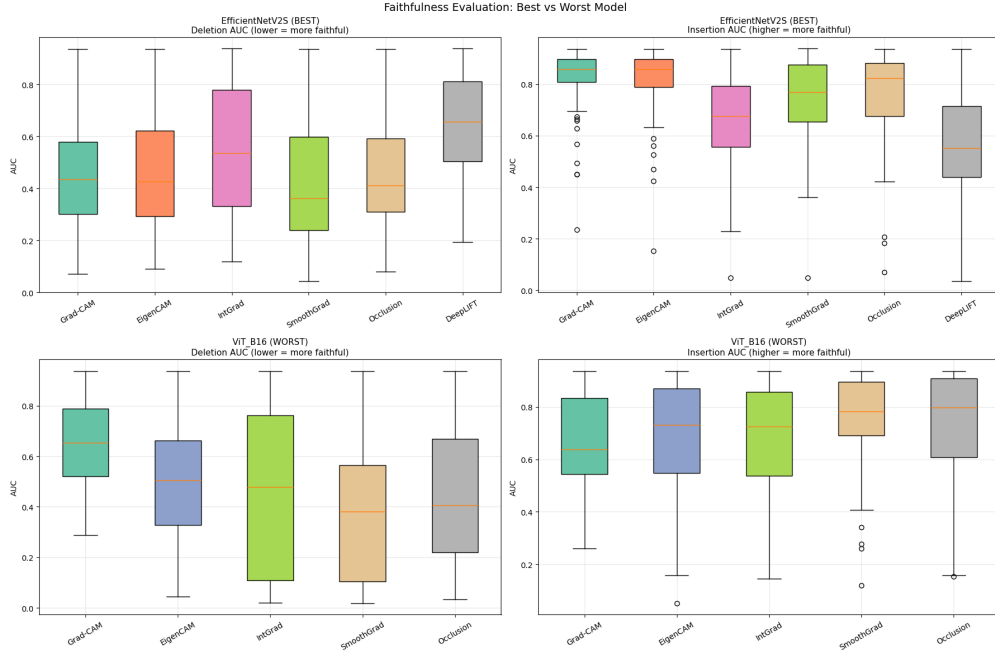


Figure 2: Barplots for the faithfulness evaluation of the best and worst model.

Figure 2 highlights key distributional patterns. On EfficientNetV2-S, Grad-CAM and EigenCAM show the tightest interquartile ranges, indicating consistent faithfulness across images. DeepLIFT’s insertion and deletion distributions largely overlap, confirming it attributes importance to irrelevant regions. On ViT-B/16, Grad-CAM’s deletion AUC shifts from a median of ~ 0.42 to ~ 0.65 —removing its top-attributed pixels barely affects ViT predictions—while SmoothGrad’s distributions remain stable across architectures, explaining its architecture-robust faithfulness.

5.2 CASCADING RANDOMIZATION / SANITY CHECK (RQ2)

Following the sanity check protocol of Adebayo et al. (2018), we progressively randomize ResNet50 weights from the output layer to the input layer and measure SSIM and Spearman $|\rho|$ between the attribution on the randomized model and the trained-model reference (Figure 3).

After full randomization, Grad-CAM and IntGrad retain $\text{SSIM} = 1.0$ with their trained-model baselines—their attributions are entirely architecture-determined, driven by skip connections and pooling geometry rather than learned features. SmoothGrad drops to $\text{SSIM} = 0.555$, making it the only tested method with partial model dependence. DeepLIFT was excluded due to a Captum incompatibility with ResNet’s reused ReLU modules. This confirms the classic finding of Adebayo et al. (2018): high deletion/insertion faithfulness (§5.1) does not guarantee that the explanation depends on the learned weights.

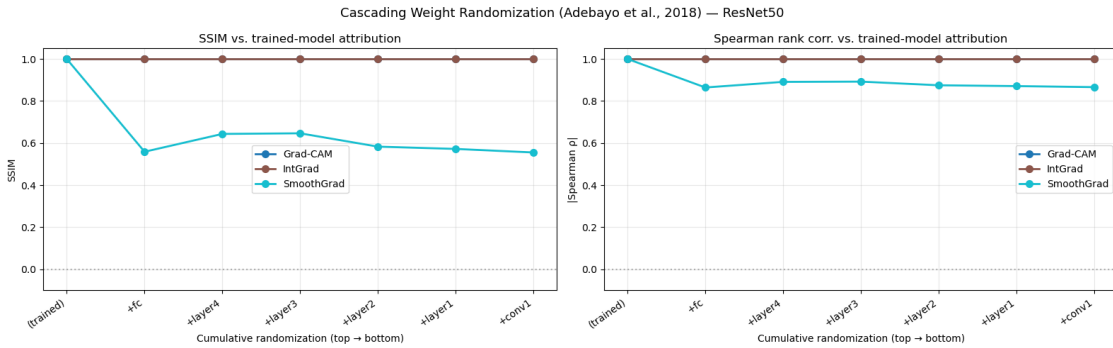


Figure 3: Cascading weight randomization on ResNet50. A faithful method should show decreasing SSIM and $|\rho|$ as more layers are randomized. SmoothGrad is the only method with partial model dependence (final $\text{SSIM} = 0.555$); Grad-CAM and IntGrad remain at $\text{SSIM} = 1.0$ throughout.

5.3 TCAV (RQ3)

Both EfficientNetV2-S and ViT-B/16 produce saturated class-vs-class TCAV scores: diagonal ≈ 1.00 , off-diagonal ≈ 0.00 , all statistically significant ($p < 0.01$). This confirms both models have fully separated class representations in penultimate space, but TCAV cannot discriminate between models at this accuracy ceiling. The shortcut-detection probe (brightness, contrast, symmetry) also saturates, limiting its diagnostic value. We attribute this to the binary sign-test nature of TCAV, which loses resolution when accuracy approaches 100%.

5.4 STABILITY (RQ4)

Under Gaussian noise ($\sigma=0.05$), Grad-CAM is the most stable (SSIM = 0.708), followed by Occlusion (0.410) and IntGrad (0.386). SmoothGrad is the least stable (SSIM = 0.124), as its noise-averaging mechanism compounds with input noise rather than canceling it. This creates a fundamental tension with its model-dependence advantage.

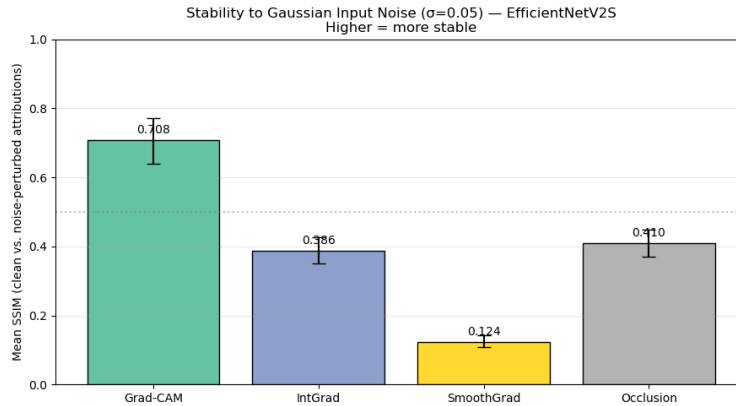


Figure 4: Stability under Gaussian input noise ($\sigma=0.05$) on EfficientNetV2-S. Bars show mean SSIM between clean and perturbed attributions (higher = more stable); error bars are bootstrap 95% CIs.

Figure 4 reveals a wide stability gap between methods. Grad-CAM is the most stable (SSIM = 0.708), benefiting from its coarse spatial resolution which smooths out input noise. SmoothGrad is the least stable (SSIM = 0.124)—its noise-averaging mechanism compounds with input perturbation rather than canceling it. Combined with its model-dependence advantage from §5.2, this creates a faithfulness–stability trade-off: the most model-dependent method is also the least reproducible across near-identical scans.

5.5 INTER-METHOD AGREEMENT (RQ5)

Grad-CAM and EigenCAM are strongly correlated ($\rho = +0.89$), operating on the same feature maps with different channel-weighting strategies. IntGrad and SmoothGrad show low correlation with the CAM cluster ($\rho \approx 0.09$ – 0.16) and with each other ($\rho \approx 0.09$), confirming they capture different aspects of the model’s reasoning. The methods are not redundant.

Figure 5 reveals two distinct clusters. Grad-CAM and EigenCAM are strongly correlated ($\rho = 0.89$), as both operate on the same convolutional feature maps and differ only in channel weighting. Occlusion shows moderate agreement with this CAM cluster ($\rho \approx 0.55$ – 0.58). In contrast, IntGrad is weakly correlated with every other method ($\rho \leq 0.16$), suggesting it captures a fundamentally different signal. The low cross-family correlations confirm the methods are complementary rather than redundant, supporting the use of multiple XAI methods in clinical practice.

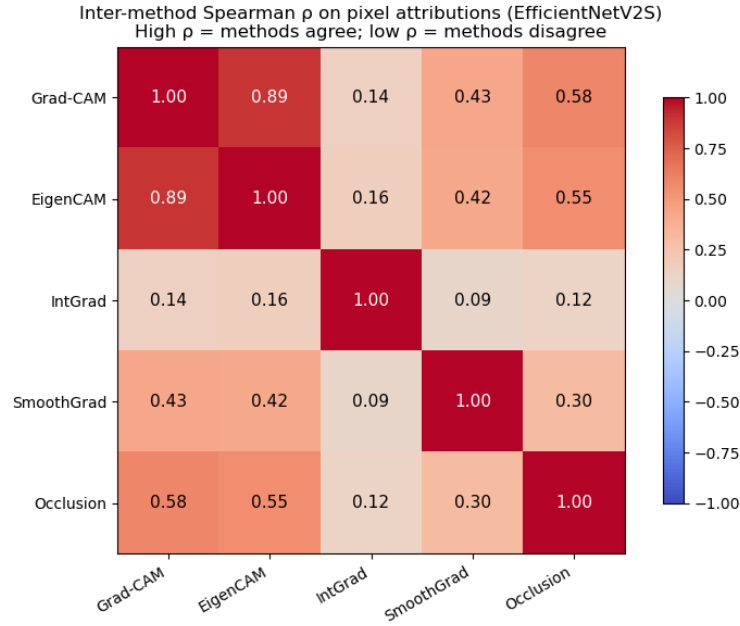


Figure 5: Inter-method Spearman ρ on pixel-level attributions, averaged over 20 images on EfficientNetV2-S. Higher values indicate methods that identify similar important regions.

5.6 COST VS. FAITHFULNESS PARETO (RQ6)

Grad-CAM (52 ms, $\Delta = +0.364$) is the sole Pareto-optimal method: no other method is both faster *and* more faithful. EigenCAM (173 ms) is dominated. SmoothGrad (3.14 s) and Occlusion (17.8 s) offer marginal faithfulness gains at $60\times$ and $340\times$ the cost, respectively.

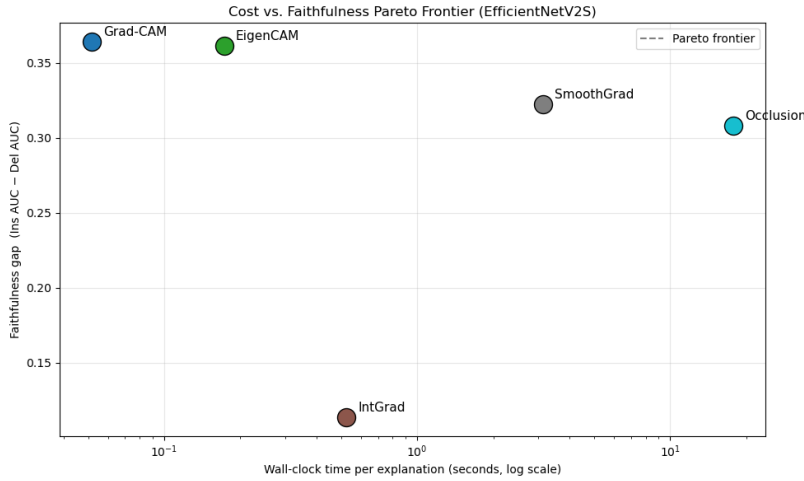


Figure 6: Cost vs. faithfulness Pareto frontier on EfficientNetV2-S. Each point represents a method; the dashed line connects non-dominated methods. The x-axis is log-scaled.

Figure 6 shows that Grad-CAM is the sole Pareto-optimal method: at 52 ms per explanation, it is $3\times$ faster than EigenCAM, $60\times$ faster than SmoothGrad, and $340\times$ faster than Occlusion, with equal or higher faithfulness. IntGrad is dominated on both axes—slower than Grad-CAM yet substantially less faithful. For real-time clinical decision support (<100 ms budget), Grad-CAM is the only viable option.

5.7 EXEMPLAR-BASED EXPLANATIONS

Beyond pixel-level attributions, we provide example-based explanations by retrieving the 5 nearest training neighbours in penultimate-layer cosine similarity for each query image. This answers a complementary question: “what training examples does this image resemble from the model’s internal perspective?”

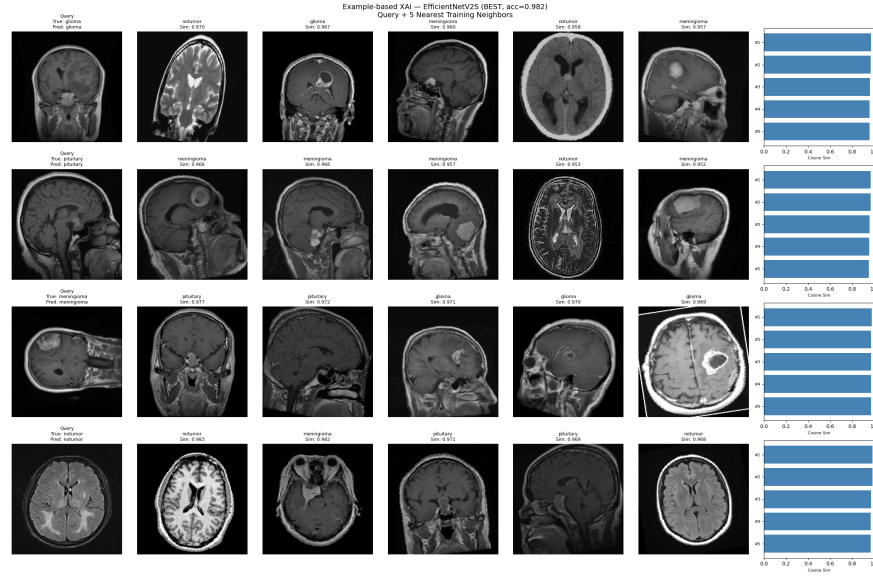


Figure 7: Exemplar-based XAI on EfficientNetV2-S (best model, acc=0.982): for each query (left), the 5 nearest training neighbours are shown with cosine similarity bars (right).

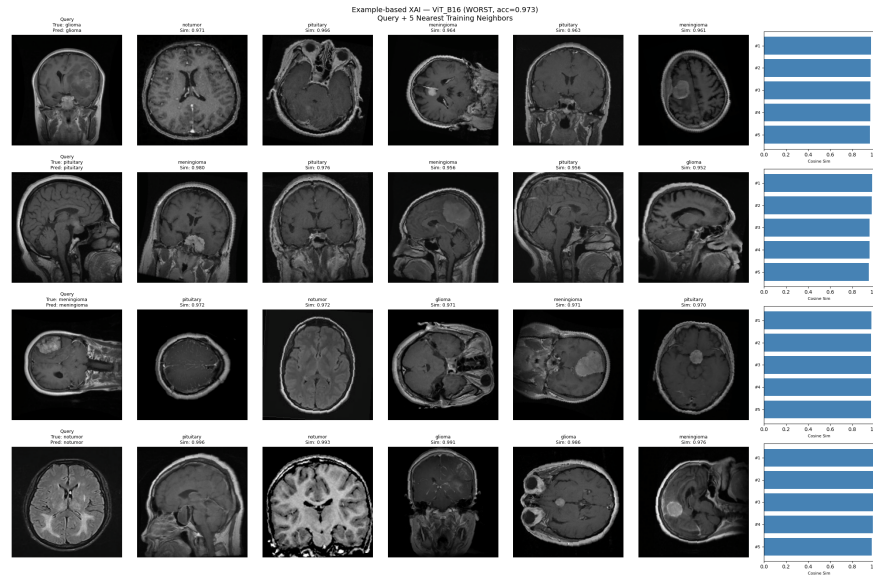


Figure 8: Same protocol on ViT-B/16 (worst model, acc=0.973).

Figures 7–8 show that both models retrieve mixed-class neighbours despite correct predictions. EfficientNetV2-S returns neighbours with similarity >0.90 across different classes, suggesting its embeddings capture shared low-level features. ViT-B/16 shows even higher similarities (>0.96) with similarly mixed neighbours, indicating a more compressed embedding space. This highlights that high classification accuracy does not imply class-coherent embeddings, and that exemplar-based explanations can surface representation weaknesses invisible to standard accuracy metrics.

6 CONCLUSIONS

Key findings. (1) Faithfulness is method- and architecture-dependent: Grad-CAM leads on CNNs, SmoothGrad leads on ViT. (2) Cascading randomization reveals that Grad-CAM and IntGrad are architecture-determined ($\text{SSIM} = 1.0$ after full randomization); SmoothGrad is the only partially model-dependent method. (3) TCAV saturates on this task (diagonal ≈ 1.00 , off-diagonal ≈ 0.00), confirming strong class separation but providing no discriminative power between models. (4) Stability and model dependence are inversely related: the most faithful method (Grad-CAM) is the most stable but architecture-dependent; the most model-dependent method (SmoothGrad) is the least stable. (5) Methods cluster into CAM and gradient families with low cross-family agreement, confirming they are complementary rather than redundant. (6) Grad-CAM is the only Pareto-optimal method on the cost–faithfulness frontier.

Recommendation. For clinical deployment, we recommend presenting **Grad-CAM** (fast, stable, spatially localized) and **SmoothGrad** (pixel-level detail, partially model-dependent) side-by-side. Disagreement between them should flag a prediction for clinician review.

Limitations. (1) No ground-truth tumor masks precluded pointing-game metrics. (2) TCAV saturates at high accuracy; continuous sensitivity measures would be more informative. (3) Cascading randomization was limited to ResNet50. (4) DeepLIFT was used instead of proper LRP. (5) Single-institution dataset.

Future work. (1) Extend to mask-annotated datasets (e.g., BraTS) for localization metrics. (2) Replace DeepLIFT with Zennit-LRP. (3) Design radiologist-derived concept sets for clinical TCAV. (4) User study with radiologists. (5) Repeat on harder tasks (tumor grading) where TCAV may be more informative.

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